



CHEMISTRY CH.8 — CHEMICAL REACTIONS — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this chapter:

1. Types of Chemical Reactions: Overview
2. Combination Reactions
3. Decomposition Reactions
4. Displacement Reactions & Reactivity Series
5. Double Displacement & Precipitation
6. Neutralization Reactions
7. Redox Reactions: Oxidation & Reduction
8. Exothermic vs Endothermic Reactions
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10. Important Industrial & Daily Life Reactions
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1. Types of Chemical Reactions: Overview

- Combination ($A + B \rightarrow AB$): Do ya zyada reactants milke ek product. Examples: $2H_2 + O_2 \rightarrow 2H_2O$; $N_2 + 3H_2 \rightarrow 2NH_3$; $H_2 + Cl_2 \rightarrow 2HCl$; $H_2 + Br_2 \rightarrow 2HBr$; $SO_3 + H_2O \rightarrow H_2SO_4$; $CaO + H_2O \rightarrow Ca(OH)_2$; $2Mg + O_2 \rightarrow 2MgO$; $2Cu + O_2 \rightarrow 2CuO$; $C + O_2 \rightarrow CO_2$; $2CO + O_2 \rightarrow 2CO_2$. General form $A + B \rightarrow AB$ (do ya zyada compounds mil ke ek single naya compound banate hain). [Q1002, Q1003, Q1025, Q1062, Q1074, Q1248, Q1285, Q1309, Q1326, Q1329, Q1052, Q1117, Q1123, Q1160, Q1163, Q1178, Q1188, Q1199]
- Decomposition ($AB \rightarrow A + B$): Ek reactant toot ke simpler products. Types: Thermal, Electrolytic, Photolytic. AgCl sunlight mein grey ho jaata hai (decompose hoke Ag + Cl₂ banata hai). CaCO₃ heat karne pe Quick lime (CaO) + CO₂ deta hai. Heating of calcium carbonate combination reaction NAHI hai — ye decomposition hai. [Q1015, Q1020, Q1024, Q1046, Q1056, Q1060, Q1101, Q1211, Q1219, Q1253, Q1270, Q1272, Q1285, Q1311, Q1316, Q1334, Q1124, Q1137, Q1142, Q1147]
- Displacement ($A + BC \rightarrow AC + B$): Zyada reactive element kam reactive element ko displace karta hai. [Q1018, Q1026, Q1027, Q1031, Q1043, Q1049, Q1050, Q1060, Q1065, Q1069, Q1080, Q1095, Q1099, Q1215, Q1234, Q1259, Q1264, Q1294, Q1306, Q1325, Q1116, Q1121, Q1125, Q1126, Q1150, Q1154, Q1159, Q1164, Q1172, Q1176, Q1191, Q1192, Q1193, Q1196, Q1291]
- Double Displacement ($AB + CD \rightarrow AD + CB$): Dono reactants ke ions exchange hote hain. Precipitation aur neutralization isi ke under aate hain — "double substitution reaction" bhi kehte hain (do groups/ions simultaneously exchange hote hain). [Q997, Q1001, Q1011, Q1039, Q1041, Q1057, Q1058, Q1063, Q1075, Q1076, Q1077, Q1084, Q1085, Q1104, Q1108, Q1217, Q1222, Q1229, Q1237, Q1249, Q1251, Q1257, Q1263, Q1296, Q1307, Q1323, Q1122, Q1127, Q1129, Q1139, Q1152, Q1167, Q1191]
- Neutralization (Acid + Base $\rightarrow$ Salt + Water): $H^+ + OH^- \rightarrow H_2O$. Exothermic hota hai. NaOH (base) + CH₃COOH (acid) $\rightarrow$ CH₃COONa + H₂O bhi neutralization ka example hai (weak acid ke saath). $Ca(OH)_2$ + conc. H₂SO₄ $\rightarrow$ CaSO₄ + H₂O bhi exothermic neutralization hai. [Q1007, Q1012, Q1013, Q1014, Q1016, Q1017, Q1023, Q1076, Q1113, Q1114, Q1238, Q1301, Q1119, Q1133, Q1141, Q1170, Q1197, Q1040]
- Precipitation: Jab insoluble solid (precipitate) aqueous solution mein banta hai — cation ek reactant ka aur anion dusre reactant ka combine hoke insoluble ionic solid banate hain. [Q998, Q1047, Q1067, Q1072, Q1073, Q1084, Q1085, Q1089, Q1092, Q1100, Q1102, Q1104, Q1108, Q1110, Q1217, Q1229, Q1251, Q1257, Q1263, Q1323, Q1120, Q1127, Q1129, Q1130, Q1135, Q1145, Q1146, Q1148, Q1152, Q1165, Q1168, Q1171, Q1175, Q1180, Q1183, Q1185, Q1189, Q1190, Q1194]
- Redox Reaction: Oxidation aur reduction simultaneously. [Q997, Q1006, Q1008, Q1009, Q1022, Q1031, Q1032, Q1034, Q1042, Q1059, Q1065, Q1069, Q1082, Q1090, Q1094, Q1096, Q1103, Q1204, Q1220, Q1224, Q1228, Q1230, Q1233, Q1235, Q1236, Q1252, Q1256, Q1269, Q1273, Q1299, Q1314, Q1318, Q1321, Q1083, Q1132, Q1134, Q1144, Q1149, Q1155, Q1161, Q1166, Q1169, Q1173, Q1177, Q1187, Q1195, Q1196]

- Combustion: Substance burns in O_2 ; heat/light release hoti hai. Complete combustion $\rightarrow CO_2 + H_2O$; incomplete $\rightarrow CO/C$. Coal aur petroleum ka combustion Nitrogen aur Sulphur ke harmful oxides (NO_x , SO_x) bhi banata hai — ye acid rain/smog cause karte hain. Iron bhi air mein burn karke Fe_2O_3 banata hai ($4Fe + 3O_2 \rightarrow 2Fe_2O_3$). [Q1044, Q1068, Q1213, Q1246, Q1266, Q1326, Q1292, Q1245]
- Addition: Molecules combine to form larger one; hydrogenation of oils (Ni catalyst). [Q1010, Q1216]
- Substitution: Functional group replace hota hai; $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$. Sunlight mein hydrocarbons ke saath Cl_2 rapidly add/substitute hoti hai (halogenation). [Q1038, Q1279]

2. Combination Reactions

- Hydrogen + Oxygen $\rightarrow$ Water ($2H_2 + O_2 \rightarrow 2H_2O$). [Q1002]
- Nitrogen + Hydrogen $\rightarrow$ Ammonia ($N_2 + 3H_2 \rightarrow 2NH_3$). [Q1021, Q1025]
- Sulphur trioxide + Water $\rightarrow$ Sulphuric acid. [Q1062]
- Calcium oxide + Water $\rightarrow$ Calcium hydroxide (slaked lime); ye exothermic hai. [Q1029, Q1111, Q1274, Q1278, Q1309, Q1329, Q1118, Q1123, Q1163]
- Carbon monoxide + Hydrogen $\rightarrow$ Methanol ($CO + 2H_2 \rightarrow CH_3OH$ at 340 atm). [Q1061, Q1088]
- $Mg + O_2 \rightarrow 2MgO$; $Cu + O_2 \rightarrow 2CuO$; $C + O_2 \rightarrow CO_2$. Balanced form: $2Mg(s) + O_2(g) \rightarrow 2MgO(s)$. [Q1248, Q1285, Q1329, Q1117]
- Hydrogen + Bromine $\rightarrow$ Hydrogen Bromide: $H_2 + Br_2 \rightarrow 2HBr$ (balanced equation). [Q1052]

3. Decomposition Reactions

- Thermal Decomposition: Heat se compound toot ta hai.
 - $ZnCO_3 \rightarrow ZnO + CO_2$. [Q1015]
 - $CaCO_3$ (limestone) $\rightarrow$ CaO (Quick lime) + CO_2 . [Q1048, Q1211, Q1137, Q1147]
 - $2KClO_3 \rightarrow 2KCl + 3O_2$. [Q1024]
 - $2FeSO_4 \rightarrow Fe_2O_3 + SO_2 + SO_3$. [Q1056, Q1101, Q1334]
 - $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$ (brown fumes of NO_2). [Q1036, Q1219]
 - $NH_4NO_3 \rightarrow N_2O + 2H_2O$. [Q1285]
 - $NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$ (baking soda). [Q1320]
 - Heating of calcium carbonate ($CaCO_3$) is a decomposition reaction, NOT a combination reaction. [Q1142]
- Electrolytic Decomposition: Electricity se.
 - $2H_2O \rightarrow 2H_2 + O_2$ (electrolysis of water). [Q1024, Q1232]
 - $2NaCl(\text{molten}) \rightarrow 2Na + Cl_2$. [Q1024]
- Photolytic Decomposition: Light se.
 - $2AgCl \rightarrow 2Ag + Cl_2$ (sunlight); AgCl grey ho jaata hai sunlight mein decompose hoke. [Q1020, Q1037, Q1253, Q1124]
 - $2AgBr \rightarrow 2Ag + Br_2$ (sunlight). [Q1024, Q1070]
- ⚠ Trap: $AgCl/AgBr$ sirf sunlight mein decompose hote hain, heat se nahi — ye photolytic hain. [Q1020, Q1037, Q1070, Q1253]

4. Displacement Reactions & Reactivity Series

- Reactivity series: $K > Na > Ca > Mg > Al > Zn > Fe > Pb > H > Cu > Hg > Ag > Pt > Au$. [Q1077, Q1304]

- ECS (Electrochemical Series): $\text{Li} > \text{K} > \text{Ba} > \text{Ca} > \text{Na} > \text{Mg} > \text{Al} > \text{Mn} > \text{Zn} > \text{Cr} > \text{Fe} > \text{Co} > \text{Ni} > \text{Sn} > \text{Pb} > \text{H} > \text{Cu} > \text{Hg} > \text{Ag} > \text{Pt} > \text{Au}$. [Q1095]
- Zyada reactive metal kam reactive metal ke salt solution ko displace kar sakta hai. [Q1018, Q1026, Q1031, Q1095, Q1099, Q1125, Q1192]
- Examples:
 - $\text{Mg} + \text{CuSO}_4 \rightarrow \text{MgSO}_4 + \text{Cu}$ (blue colour fades, red-brown deposit). [Q1026, Q1031, Q1068]
 - $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$. [Q1043, Q1049, Q1099, Q1264, Q1154, Q1193]
 - $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$ (iron ne copper ko displace kiya). [Q1041, Q1049, Q1215, Q1121, Q1125]
 - $\text{Pb} + \text{CuCl}_2 \rightarrow \text{PbCl}_2 + \text{Cu}$ (ismein sirf Cl atom hi ion ke roop mein rehta hai). [Q1041, Q1060, Q1325, Q1164]
 - $\text{Ca} + \text{CuSO}_4 \rightarrow \text{CaSO}_4 + \text{Cu}$. [Q1018]
 - $\text{Al} + \text{CuSO}_4 \rightarrow \text{Al}_2(\text{SO}_4)_3 + \text{Cu}$. [Q1080]
 - $\text{Zn} + \text{FeCl}_2 \rightarrow \text{ZnCl}_2 + \text{Fe}$. [Q1325]
 - $2\text{Al} + \text{Fe}_2\text{O}_3 \rightarrow \text{Al}_2\text{O}_3 + 2\text{Fe}$ (Aluminium Iron se zyada reactive hai, isliye displace karta hai). [Q1116, Q1196]
 - $2\text{KI} + \text{Cl}_2 \rightarrow 2\text{KCl} + \text{I}_2$ (halogen displacement). [Q1027]
 - $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2 \uparrow$ (H_2 combustible, popping sound). [Q1112]
 - $\text{Zn} + \text{H}_2\text{SO}_4 \rightarrow \text{ZnSO}_4 + \text{H}_2$. [Q1091, Q1176]
 - $\text{Fe} + 2\text{HCl} \rightarrow \text{FeCl}_2 + \text{H}_2$ (iron filings + dilute HCl). [Q1291]
 - $\text{Zn} + \text{NaOH} \rightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2$ (sodium zincate). [Q1234]
 - $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$ (H_2O oxidising agent, coefficients $x=3, y=4, z=4$). [Q1126, Q1150]
 - $\text{Zn} + 2\text{AgNO}_3 \rightarrow \text{Zn}(\text{NO}_3)_2 + 2\text{Ag}$ (zinc silver nitrate se silver displace karta hai). [Q1159]
 - Element A jo chlorine ke saath high-melting, water-soluble ACl banata hai — A = Sodium ($2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$). [Q1172]
- Platinum aur Silver CuSO_4 se Cu displace nahi kar sakte (kam reactive). [Q1026]
- H_2 gas dilute HCl se sirf woh metals liberate kar sakti hain jo H se upar hain: Pb kar sakta hai; Cu, Hg, Ag nahi kar sakte. [Q1098, Q1271, Q1304]
- $\text{Cu} + \text{PbCl}_2 \rightarrow \text{CuCl}_2 + \text{Pb}$ NOT possible (Cu Pb se neeche hai ECS mein). [Q1095]
- $\text{Cu} + \text{ZnSO}_4 \rightarrow \text{NO reaction}$ (Cu Zn se kam reactive). [Q1262]
- Double substitution/displacement reaction mein do groups (anions ya cations) simultaneously exchange hote hain — displacement se alag hai jismein ek single element replace hota hai. [Q1191]
- ⚠ Trap: $\text{AgBr} \rightarrow \text{Ag} + \text{Br}_2$ displacement nahi hai, ye photolytic decomposition hai. [Q1060]

5. Double Displacement & Precipitation

- General form: $\text{AB} + \text{CD} \rightarrow \text{AD} + \text{CB}$. [Q1039, Q1041, Q1058, Q1085, Q1104, Q1108, Q1122, Q1152]
- Examples:
 - $\text{AgNO}_3 + \text{NaCl} \rightarrow \text{AgCl} \downarrow + \text{NaNO}_3$ (white precipitate). [Q997, Q1001, Q1011, Q1072, Q1122, Q1129]
 - $\text{AgNO}_3 + \text{KBr} \rightarrow \text{AgBr} + \text{KNO}_3$. [Q1001]
 - $\text{Na}_2\text{SO}_4 + \text{BaCl}_2 \rightarrow \text{BaSO}_4 \downarrow + 2\text{NaCl}$ (white precipitate). [Q1041, Q1057, Q1084, Q1085, Q1104, Q1152, Q1165, Q1171, Q1194]
 - $\text{CuSO}_4 + \text{BaCl}_2 \rightarrow \text{BaSO}_4 \downarrow + \text{CuCl}_2$. [Q1047]
 - $\text{CuSO}_4 + \text{H}_2\text{S} \rightarrow \text{CuS} \downarrow + \text{H}_2\text{SO}_4$. [Q1077, Q1183]
 - $\text{Pb}(\text{NO}_3)_2 + 2\text{KI} \rightarrow \text{PbI}_2 \downarrow + 2\text{KNO}_3$ (yellow precipitate). [Q1067, Q1092, Q1102, Q1108, Q1110, Q1120, Q1146]
 - $2\text{NaOH} + \text{CuSO}_4 \rightarrow \text{Cu}(\text{OH})_2 \downarrow + \text{Na}_2\text{SO}_4$. [Q1075]
 - $\text{CuO} + 2\text{HCl} \rightarrow \text{CuCl}_2 + \text{H}_2\text{O}$ (blue-green colour CuCl_2 ki wajah se). [Q1087, Q1157]

- $\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2\uparrow$ (effervescence, gas lime water milky karta hai). [Q1251, Q1323, Q1136]
- $\text{CaCl}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + 2\text{HCl}$. [Q1097]
- Marble chips (CaCO_3) + dilute $\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2$. [Q1257, Q1200]
- $\text{Na}_2\text{S} + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{S}\uparrow$ (rotten-egg smell wali gas). [Q1051]
- $2\text{KBr} + \text{BaI}_2 \rightarrow 2\text{KI} + \text{BaBr}_2$ (BaBr_2 insoluble hai). [Q1145]
- $3\text{BaCl}_2 + \text{Al}_2(\text{SO}_4)_3 \rightarrow 3\text{BaSO}_4\downarrow + 2\text{AlCl}_3$. [Q1135, Q1168]
- $\text{Na}_2\text{CO}_3 + \text{CaCl}_2 \rightarrow \text{CaCO}_3\downarrow + 2\text{NaCl}$ (double displacement + precipitation dono). [Q1185, Q1189]
- $\text{Ca}(\text{OH})_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$ (left side mein 4 oxygen atoms hote hain). [Q1190]
- BaSO_4 : water insoluble white crystalline solid; odourless. Uses: X-ray contrast agent (barium meal), oil drilling fluid, plastic filler. [Q1084, Q1089]
- Precipitate ki physical state hamesha solid hoti hai; precipitation reaction hamesha ek insoluble ionic salt banati hai. [Q1100, Q1127, Q1130, Q1175]
- Double displacement reactions ke 3 typical signs: precipitate banna, gas evolve hona, colour change hona — teeno hi possible hain. [Q1148]
- Precipitation reactions double displacement reactions ki hi ek specific class hain (neutralization bhi). [Q1180]
- Ionic compounds ke beech hi double displacement reaction hoti hai (ions exchange). [Q1167]
- Lemon juice milk mein daalne se curdling hoti hai — ye bhi ek precipitation-type reaction hai (lactic acid forms, paneer banta hai). [Q1189]
- ⚠ Trap: $\text{AgNO}_3 + \text{NaCl} \rightarrow \text{AgCl} + \text{NaNO}_3$ redox nahi hai — ye double displacement hai. Redox mein oxidation number change hota hai. [Q997]

6. Neutralization Reactions

- Acid + Base $\rightarrow$ Salt + Water. [Q1007, Q1012, Q1014, Q1016, Q1017, Q1023, Q1076, Q1113, Q1133]
- Products hamesha H_2O + Salt hote hain. [Q1113]
- Examples:
 - $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$. [Q1017, Q1023, Q1076]
 - $\text{HNO}_3 + \text{KOH} \rightarrow \text{KNO}_3 + \text{H}_2\text{O}$. [Q1014, Q1017, Q1114]
 - $\text{Ca}(\text{OH})_2 + 2\text{HCl} \rightarrow \text{CaCl}_2 + 2\text{H}_2\text{O}$. [Q1012]
 - $2\text{LiOH} + \text{H}_2\text{SO}_4 \rightarrow \text{Li}_2\text{SO}_4 + 2\text{H}_2\text{O}$. [Q1016]
 - $\text{Ca}(\text{OH})_2 + \text{H}_2\text{CO}_3 \rightarrow \text{CaCO}_3 + 2\text{H}_2\text{O}$. [Q1114]
 - $\text{H}_3\text{PO}_4 + 3\text{KOH} \rightarrow \text{K}_3\text{PO}_4 + 3\text{H}_2\text{O}$. [Q1114]
 - $\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$. [Q1238]
 - $\text{CaO} + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O}$. [Q1301]
 - $\text{NaOH} + \text{CH}_3\text{COOH} \rightarrow \text{CH}_3\text{COONa} (\text{X}) + \text{H}_2\text{O} (\text{Y})$ — weak acid (acetic acid) + strong base ka neutralization. [Q1119, Q1170]
 - $2\text{NaOH} + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$. [Q1141]
 - $2\text{HNO}_3 + \text{Ca}(\text{OH})_2 \rightarrow \text{Ca}(\text{NO}_3)_2 + 2\text{H}_2\text{O}$ (balancing coefficients a=2, b=1, c=2). [Q1197]
 - $\text{Ca}(\text{OH})_2 + \text{H}_2\text{SO}_4(\text{conc.}) \rightarrow \text{CaSO}_4 + 2\text{H}_2\text{O}$ (bahut exothermic; energy itni release hoti hai ki fire bhi cause kar sakta hai). [Q1040]
- Exothermic reaction hai. [Q1012, Q1029, Q1111]
- 5 ml NaOH + 5 ml HCl neutralize; toh 10 ml NaOH ke liye 10 ml HCl chahiye (1:1 ratio). [Q1023]
- Antacids mein $\text{Mg}(\text{OH})_2$ ya MgCO_3 use hote hain excess HCl neutralize karne ke liye. [Q1012]
- ⚠ Trap: $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca}(\text{OH})_2$ neutralization nahi hai — ye combination + exothermic reaction hai. [Q1114]

7. Redox Reactions: Oxidation & Reduction

- Oxidation: O_2 gain, H_2 loss, electrons loss. Substance jab O_2 ke saath combine hota hai toh woh oxidize hota hai. [Q1022, Q1034, Q1082, Q1090, Q1204, Q1235, Q1083, Q1149]
- Reduction: H_2 gain, O_2 loss, electrons gain. "Loss of electrons" reduction ka INCORRECT statement hai — ye toh oxidation ki definition hai. [Q1032, Q1034, Q1096, Q1233, Q1236, Q1166, Q1177]
- Oxidizing agent: Dusre ko oxidize karta hai (khud reduce hota hai). [Q1022]
- Reducing agent: Dusre ko reduce karta hai (khud oxidize hota hai). [Q1022]
- Examples:
 - $ZnO + C \rightarrow Zn + CO$: ZnO reduced (oxidizing agent), C oxidized (reducing agent). [Q1008, Q1042, Q1069, Q1094, Q1132, Q1155]
 - $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$: Fe_2O_3 reduced, CO oxidized (reducing agent). [Q1009, Q1256]
 - $2Al + Fe_2O_3 \rightarrow Al_2O_3 + 2Fe$: Fe_2O_3 oxidizing agent (reduced), Al reducing agent (oxidized). [Q1006, Q1196]
 - $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$: MnO_2 reduced (oxidizing agent), HCl oxidized to Cl_2 — " Cl_2 oxidized to HCl" statement INCORRECT hai (ulta hota hai). [Q997, Q1059, Q1134]
 - $CuO + H_2 \rightarrow Cu + H_2O$: CuO reduced, H_2 oxidized. [Q1032, Q1065, Q1233, Q1236, Q1252, Q1321]
 - $CuSO_4 + Mg \rightarrow MgSO_4 + Cu$: Mg reducing agent. [Q1050]
 - $Cu + \text{conc. } HNO_3 \rightarrow Cu(NO_3)_2 + NO_2 + H_2O$: Cu oxidized, N reduced. [Q1103]
 - $MgO + CO \rightarrow Mg + CO_2$: CO oxidized, MgO reduced. [Q1224]
 - $PbO_2 + 4HCl \rightarrow PbCl_2 + 2H_2O + Cl_2$: HCl oxidized ($Cl^- \rightarrow Cl_2$). [Q1228]
 - $KMnO_4 + FeSO_4$ (dilute H_2SO_4 mein): $KMnO_4$ strong oxidizing agent — $Fe^{+2} \rightarrow Fe^{+3}$; purple colour fades. [Q1230]
 - $2PbO + C \rightarrow 2Pb + CO_2$: C oxidized, PbO reduced ($x=2, y=2, z=1$). [Q1269, Q1161]
 - $SnCl_2 + 2HgCl_2 \rightarrow Hg_2Cl_2 + SnCl_4$: $SnCl_2$ oxidized ($Sn^{+2} \rightarrow Sn^{+4}$). [Q1318]
 - $H_2O_2 + Cl_2 \rightarrow 2HCl + O_2$: H_2O_2 reducing agent ($O^{-1} \rightarrow O^0$). [Q1314]
 - $2Cu + O_2 \rightarrow 2CuO$: Copper aur Oxygen dono involved reaction mein — Cu oxidize hota hai (loses electrons). [Q1173]
 - $Zn \rightarrow Zn^{2+}$ formation bhi oxidation hai (electron loss). [Q1169]
 - S (Sulphide ion) sulphur se banta hai reduction ke through. [Q1144]
- Ethanol $\rightarrow$ Ethanoic acid (CH_3COOH) oxidation hai (alkaline $KMnO_4$ / acidified $K_2Cr_2O_7$ + heat). [Q1019, Q1195]
- Rancidity = oxidation of fats/oils. Chips bags mein N_2 flush ki jaati hai oxidation rokne ke liye. [Q1054, Q1298, Q1313]
- Rusting = oxidation of iron. [Q1096]
- Redox reaction ki definition: oxidation aur reduction simultaneously ek hi reaction mein hote hain (electron transfer). [Q1187]
- Mg ribbon burning air mein (oxidation reaction) $\rightarrow$ MgO forms. [Q1149, Q1156, Q1140]
- ⚠ Trap: $2PbO + C \rightarrow 2Pb + CO_2$ mein " CO_2 oxidized" aur " Pb reduced" statements galat hain — actually C oxidized hota hai aur PbO reduced. [Q1269]
- ⚠ Trap: $MnO_2 + 4HCl$ mein MnO_2 oxidizing agent hai (reduce hota hai $Mn^{+4} \rightarrow Mn^{+2}$), HCl reducing agent (oxidize hoke Cl_2 banata hai). [Q1059]

8. Exothermic vs Endothermic Reactions

- Exothermic: Heat release hoti hai. ΔH negative. Examples: combustion, neutralization, respiration, rusting, slaking of lime, freezing of water, dissolving acid/alkali in water, decomposition of vegetable matter, burning of natural gas. [Q1029, Q1109, Q1111, Q1213, Q1241, Q1254, Q1255, Q1280, Q1288]
 - Respiration: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{Energy}$ (exothermic reaction). [Q1029, Q1162]
 - $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O + \text{Heat}$. [Q1029, Q1213, Q1158]

- Endothermic: Heat absorb hoti hai. Examples: melting ice, melting solid salts, photosynthesis, dry ice, alkane cracking, NH_4Cl in water, evaporation, thermal decomposition of FeSO_4 . [Q1035, Q1109, Q1218, Q1288, Q1334]
- $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2 + \text{heat}$ — vigorously exothermic combination. [Q1111, Q1274, Q1278]
- ⚠ Trap: $\text{CaO} + \text{H}_2\text{O}$ mein heat release hoti hai; ye exothermic combination hai. [Q1111]
- ⚠ Trap: Dissolving acid/alkali in water exothermic hota hai — endothermic nahi. [Q1254]

9. Balancing Chemical Equations & Laws

- Law of Conservation of Mass: Matter create/destroy nahi hota — isliye balancing zaroori hai. [Q1000, Q1107, Q1212, Q1287]
- Hit and Trial Method se balance karte hain. [Q1064, Q1081]
- State symbols: (s) = solid, (l) = liquid, (g) = gas, (aq) = aqueous (dissolved in water). Triangle (Δ) heating indicate karta hai; $\rightleftharpoons$ reversible reactions ke liye. [Q1013, Q1143]
- Reaction conditions (temp, pressure, catalyst) arrow ke upar/niche likhte hain. [Q1028]
- Arrow always products ki taraf point karti hai. Reactants hamesha LHS; products RHS. [Q1055, Q1105]
- Balanced equation ka matlab: reactants aur products donon taraf har element ke atoms ki sankhya barabar. [Q1081, Q1107, Q1250, Q1297, Q1302]
- Balanced examples:
 - $3\text{BaSO}_4 + 2\text{AlCl}_3 \rightarrow 3\text{BaCl}_2 + \text{Al}_2(\text{SO}_4)_3$. [Q999]
 - $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$. [Q1030, Q1126, Q1150, Q1197]
 - $\text{S}_8 + 4\text{C} \rightarrow 4\text{CS}_2$. [Q1035]
 - $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$. [Q1021]
 - $6\text{CO}_2 + 12\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 + 6\text{H}_2\text{O}$ (Photosynthesis). [Q1045]
 - $\text{Mg}_3\text{N}_2 + 6\text{H}_2\text{O} \rightarrow 3\text{Mg(OH)}_2 + 2\text{NH}_3$. [Q1071]
 - $\text{Zn} + \text{H}_2\text{SO}_4 \rightarrow \text{ZnSO}_4 + \text{H}_2$. [Q1081]
 - $\text{CaCl}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + 2\text{HCl}$. [Q1097]
 - $\text{Cu} + 4\text{HNO}_3 \rightarrow \text{Cu(NO}_3)_2 + 2\text{NO}_2 + 2\text{H}_2\text{O}$. [Q1103]
 - $\text{SO}_2 + 2\text{H}_2\text{S} \rightarrow 3\text{S} + 2\text{H}_2\text{O}$. [Q1115]
 - $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$. [Q1297, Q1117]
 - $\text{CO}_2 + 2\text{NaOH} \rightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O}$ (CO_2 NaOH solution mein dissolve hoke sodium carbonate banata hai). [Q1186]
- Skeletal (Unbalanced) Equation: Atoms equal nahi hote, e.g., $\text{Mg} + \text{O}_2 \rightarrow \text{MgO}$. [Q1265, Q1333]
- During chemical reaction, atomic number same rehta hai (protons participate nahi karte). [Q1223]
- Galat balanced equation example: $\text{CO(g)} + \text{H}_2\text{(g)} \rightarrow \text{CH}_2\text{O(l)}$ — sahi hai $\text{CO} + \text{H}_2 \rightarrow \text{CH}_2\text{O(g)}$ (Formaldehyde), state symbol galat tha. [Q1182]
- ⚠ Trap: Balanced equation mein mass equal hota hai, volume equal nahi hoti. [Q1107]

10. Important Industrial & Daily Life Reactions

- Bleaching Powder (CaOCl_2 / $\text{Ca(OC}l)_2$): Dry slaked lime (Ca(OH)_2) + $\text{Cl}_2 \rightarrow \text{CaOCl}_2 + \text{H}_2\text{O}$. Disinfectant, bleaching agent; water ke saath Cl_2 release hoti hai. Brine electrolysis se mila Cl_2 gas hi bleaching powder banane mein use hota hai. [Q996, Q1033, Q1261, Q1331, Q1174]
- Chlor-alkali Process: $2\text{NaCl} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{Cl}_2 + \text{H}_2$ (electrolysis of brine). Anode pe oxidation: $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$ (Cl_2 gas anode pe); Cathode pe reduction: $2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{OH}^-$ (H_2 gas + NaOH cathode pe); Na^+ cathode compartment mein migrate karke NaOH banata hai. Brine electrolysis mein H_2 gas pop sound ke saath jalti hai. [Q1004, Q1093, Q1324, Q1128, Q1138, Q1153, Q1179, Q1181]

- Baking Soda (NaHCO_3): $\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2 + \text{NH}_3 \rightarrow \text{NH}_4\text{Cl} + \text{NaHCO}_3$. $\text{NaHCO}_3 + \text{H}^+ \rightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{sodium salt}$. Garam karne pe $\text{Na}_2\text{CO}_3 + \text{H}_2\text{O} + \text{CO}_2$. [Q1053, Q1320, Q1131]
- Washing Soda (Na_2CO_3): HCl ke saath CO_2 gas with brisk effervescence deta hai. Raw material = Sodium Chloride (NaCl), Solvay's process se banta hai. [Q1251, Q1323, Q1184]
- NaCl se teen important industrial compounds banate hain: Sodium Hydroxide (chlor-alkali), Baking Soda (NaHCO_3), Washing Soda (Na_2CO_3). [Q1231]
- Plaster of Paris: Calcium Sulphate Hemihydrate ($\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$); water milane pe gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) banata hai. [Q1247]
- White Washing: $\text{Ca}(\text{OH})_2$ (slaked lime) + CO_2 (air) $\rightarrow \text{CaCO}_3$ (thin white layer) + H_2O . Ye reaction 2-3 din baad whitewashed walls pe shiny finish deta hai. [Q1079, Q1275, Q1277]
- Photosynthesis: $6\text{CO}_2 + 12\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 + 6\text{H}_2\text{O}$. [Q1045]
- Slaking of lime: $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca}(\text{OH})_2 + \text{heat}$ (exothermic). [Q1029, Q1111, Q1274, Q1278]
- Ethanol + Conc. H_2SO_4 at 443 K $\rightarrow$ Ethene (C_2H_4) — dehydration. [Q1226, Q1282]
- Ethanol + $\text{Na} \rightarrow$ Sodium ethoxide + H_2 . [Q1214]
- $\text{CO} + 2\text{H}_2 \rightarrow \text{CH}_3\text{OH}$ (methanol) at 340 atm pressure. [Q1088]
- Galvanic Cell = Voltaic Cell: Chemical energy $\rightarrow$ Electrical energy. [Q1240]
- Catalyst: Reaction speed up karta hai, khud consume nahi hota. [Q1260]
- Metal oxide + Acid $\rightarrow$ Salt + Water (neutralization type). [Q1301]
- Blast furnace mein Iron oxide (Fe_2O_3) Carbon Monoxide (CO) se reduce hota hai — $2\text{C} + \text{O}_2 \rightarrow 2\text{CO}$; $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$. [Q1225]
- Sulphur (Group 16 element) vulcanized rubber banane mein use hota hai; hot concentrated HNO_3 ke saath react karke H_2SO_4 banata hai: $\text{S} + 6\text{HNO}_3 \rightarrow \text{H}_2\text{SO}_4 + 2\text{NO}_2 + 3\text{H}_2\text{O}$. [Q1303]
- Welding ke liye Ethyne (C_2H_2) + Oxygen (O_2) ka mixture use hota hai — oxy-acetylene flame bahut high temperature ki hoti hai. [Q1312]

11. Corrosion, Rancidity & Tarnishing

- Corrosion: Metals ki surface air/moisture se oxidize hoti hai. Iron $\rightarrow$ rust (red-brown, $\text{Fe}_2\text{O}_3 \cdot n\text{H}_2\text{O}$), Copper $\rightarrow$ green. Ye oxidation reaction hai. Serious problem; galvanisation (Zn coating) se prevent karte hain. Iron direct air mein burn karke bhi Fe_2O_3 banata hai (combustion, corrosion se alag). [Q1066, Q1078, Q1244, Q1281, Q1332, Q1245]
- Tarnishing:
 - Copper tarnishing: Humid air mein $\text{Cu}(\text{OH})_2 + \text{CuCO}_3 =$ green coating. [Q1239]
 - Silver tarnishing: Air mein H_2S se react karke Ag_2S (black) banata hai. [Q1203, Q1207]
- Rancidity: Fats/oils ka oxidation — taste/smell change. Rancidity rokne ke liye: airtight containers, nitrogen flushing (chips bags), antioxidants. [Q1054, Q1298, Q1313]
- Sodium (Na) itna reactive hai ki khula rakhne par oxygen ke saath explosively react karke aag pakad leta hai — isliye kerosene mein store karte hain. [Q1206]
- ⚠ Trap: Tarnishing/corrosion oxidation hai — displacement ya decomposition nahi. [Q1066, Q1078]
- ⚠ Trap: Chips bags mein N_2 flush hoti hai, O_2 nahi — oxidation prevent karne ke liye. [Q1054]

12. Physical vs Chemical Change

- Chemical Change: New substance with new properties forms — digestion, curd formation, rusting, burning, fermentation, candle burning (wax + wick). [Q1268, Q1283, Q1330]
- Physical Change: No new substance — water $\rightarrow$ vapor (evaporation), ice melting, sublimation of iodine, sugar dissolving, dissolving salt in water. [Q1268, Q1330]

- Chemical reaction hone ke indicators: gas evolution, precipitate formation, colour change, temperature change, state change — ye sab valid indicators hain. [Q1276]
- Chemical weathering rocks/soil ki molecular structure change karti hai (solution, carbonation, hydration, oxidation-reduction se) — Thermal expansion is NOT a chemical weathering process (ye purely physical hai, temperature ki wajah se expand/contract hota hai). [Q1198]
- ⚠ Trap: Water ka vapor mein badalna physical change hai, chemical nahi. [Q1268]
- ⚠ Trap: Change in size chemical reaction ka indicator nahi hai — state, colour, gas, temperature, precipitate indicators hain. [Q1208]

13. Gases, Colours & Key Observations

- CO₂: Linear covalent molecule; solid form = dry ice; water mein soluble; lime water (Ca(OH)₂) ko milky (CaCO₃) banata hai. [Q1205, Q1242, Q1317, Q1322, Q1158, Q1200]
- NO₂: Brown fumes, reddish-brown gas; smog ka colour. [Q1293, Q1319]
- Mg ribbon burns with dazzling white flame → white powder MgO. [Q1209, Q1227, Q1284, Q1286, Q1315, Q1005, Q1149, Q1156]
- AgCl: Light sensitive; dark bottle mein store karte hain kyunki sunlight mein decompose hota hai (grey ho jaata hai). [Q1253, Q1124]
- Sublimation: Solid directly gas — NH₄Cl, Iodine, Camphor, Naphthalene, Dry ice. [Q1202, Q1210, Q1290]
- H₂ isotopes: Protium (¹H), Deuterium (²H), Tritium (³H). [Q1112]
- Reversible reactions: ⇌ symbol use hota hai (e.g., N₂ + 3H₂ ⇌ 2NH₃); irreversible mein → hota hai. [Q1300, Q1305]
- Water of crystallization: Crystals heat karne pe lost hota hai → anhydrous form. CuSO₄·5H₂O (blue) → heat → anhydrous CuSO₄ (white). [Q1308, Q1310]
- Metals jab water ke saath react karte hain toh Hydrogen gas release hoti hai (metal oxide/hydroxide ke saath). Al, Fe, Zn cold/hot water se react nahi karte lekin steam se karte hain; Pb, Cu, Ag, Au bilkul react nahi karte water se. [Q1243]
- Sodium cold water ke saath violently react karta hai — H₂ gas turant aag pakad leta hai (exothermic). [Q1258]
- Calcium water ke saath Sodium se kam violently react karta hai; H₂ ke bubbles calcium ki surface se chipak jaate hain. [Q1221]
- Aluminium oxide (Al₂O₃) amphoteric oxide hai — acid aur base dono se react karta hai. [Q1267]
- Metals generally 1 se 3 (kuch mein 4, jaise Sn, Pb) valence electrons rakhte hain outermost shell mein; non-metals 4-7. [Q1289]
- ⚠ Trap: Mg₃N₂ + 6H₂O → 3Mg(OH)₂ + 2NH₃ double displacement nahi hai — ye hydrolysis type reaction hai. [Q1328]

14. Rate of Reaction

- Rate constant (k): Directly proportional to temperature; 10°C rise se rate constant doubles hota hai. [Q1201]
- Rate = k[A]^m[B]ⁿ; m, n = reaction order. [Q1201]

15. ⚠ Traps / Key Differences Summary (Full)

- ⚠ Trap: AgNO₃ + NaCl → AgCl + NaNO₃ redox nahi hai — ye double displacement hai. [Q997]
- ⚠ Trap: MnO₂ + 4HCl mein MnO₂ oxidizing agent hai (reduce hota hai Mn⁺⁴ → Mn⁺²), HCl reducing agent — "Cl₂ oxidized to HCl" ulta statement galat hai. [Q1059, Q1134]
- ⚠ Trap: Displacement mein reactivity series ka dhyan rakho — Mg Zn Fe Pb H Cu Ag Au. Ag aur Pt Cu displace

nahi kar sakte. [Q1026, Q1077]

- ⚠ Trap: $\text{Cu} + \text{PbCl}_2 \rightarrow \text{CuCl}_2 + \text{Pb}$ possible nahi — Cu Pb se neeche hai ECS mein. [Q1095]
- ⚠ Trap: H_2 liberate karne ke liye metal ko H se upar hona chahiye ECS mein; Cu, Hg, Ag nahi kar sakte. [Q1098, Q1271]
- ⚠ Trap: AgCl/AgBr ka decomposition sirf sunlight mein hota hai (photolytic), heat se nahi. [Q1020, Q1037, Q1070, Q1253]
- ⚠ Trap: Baking soda preparation ka equation: $\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2 + \text{NH}_3 \rightarrow \text{NH}_4\text{Cl} + \text{NaHCO}_3$. [Q1053]
- ⚠ Trap: Chips bags mein N_2 flush hoti hai, O_2 nahi — oxidation prevent karne ke liye. [Q1054]
- ⚠ Trap: Photosynthesis equation balance karte waqt $6\text{CO}_2 + 12\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 + 6\text{H}_2\text{O}$ hota hai. [Q1045]
- ⚠ Trap: Tarnishing/corrosion oxidation hai — displacement ya decomposition nahi. [Q1066, Q1078]
- ⚠ Trap: Ethanol $\rightarrow$ Ethanoic acid oxidation hai ($\text{KMnO}_4/\text{K}_2\text{Cr}_2\text{O}_7 + \text{heat}$). [Q1019, Q1195]
- ⚠ Trap: Neutralization hamesha exothermic hota hai. [Q1012, Q1029]
- ⚠ Trap: $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$ combination + exothermic hai, neutralization nahi. [Q1114]
- ⚠ Trap: Precipitate hamesha solid state mein hota hai, liquid nahi. [Q1100]
- ⚠ Trap: BaSO_4 water insoluble hai — isliye BaSO_4 X-ray mein safe use hota hai (Ba^{2+} toxic). [Q1089]
- ⚠ Trap: Balanced equation mein mass equal hota hai, volume equal nahi hoti. [Q1107]
- ⚠ Trap: Reactants hamesha arrow ke left side (LHS) hote hain. [Q1105]
- ⚠ Trap: Slaking of lime ($\text{CaO} + \text{H}_2\text{O}$) exothermic hai — ismein heat release hoti hai. [Q1111]
- ⚠ Trap: $\text{HI} \rightarrow \text{H}_2 + \text{I}_2$ decomposition hai — combination ka ulta. [Q1106]
- ⚠ Trap: Chlor-alkali process mein Na^+ cathode ki taraf migrate karta hai NaOH banane ke liye. [Q1093]
- ⚠ Trap: $2\text{PbO} + \text{C} \rightarrow 2\text{Pb} + \text{CO}_2$ mein " CO_2 oxidized" aur " Pb reduced" galat hain — actually C oxidized hota hai aur PbO reduced. [Q1269]
- ⚠ Trap: $\text{Cu} + \text{ZnSO}_4 \rightarrow \text{NO reaction}$ (Cu Zn se kam reactive). [Q1262]
- ⚠ Trap: Water $\rightarrow$ water vapour physical change hai, chemical nahi. [Q1268]
- ⚠ Trap: Change in size chemical reaction ka indicator nahi hai. [Q1208]
- ⚠ Trap: Dissolving acid/alkali in water exothermic hota hai. [Q1254]
- ⚠ Trap: Photosynthesis, methane chlorination, photography = photochemical; ammonia synthesis = NOT photochemical. [Q1295]
- ⚠ Trap: $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$ reversible hai, irreversible nahi. [Q1300]
- ⚠ Trap: Unbalanced equation = skeletal equation. [Q1333]
- ⚠ Trap: Anhydrous form = water of crystallization lost. [Q1308]
- ⚠ Trap: $\text{Ca} + \text{H}_2\text{O}$ ki reaction kam violent hoti hai Na se. [Q1327, Q1221, Q1258]
- ⚠ Trap: $\text{Mg}_3\text{N}_2 + 6\text{H}_2\text{O} \rightarrow 3\text{Mg(OH)}_2 + 2\text{NH}_3$ double displacement nahi hai — ye hydrolysis type hai. [Q1328]
- ⚠ Trap: "Reduction = Loss of electrons" statement galat hai — reduction mein electrons GAIN hote hain, loss oxidation mein hota hai. [Q1166]
- ⚠ Trap: Metals jo carbon se reduce ho sakte hain (Zn, Fe, Cu jaise moderately reactive) — lekin K, Na, Mg, Ca, Al jaise highly reactive metals ke oxides carbon, H_2 ya CO se reduce NAHI hote (unhe electrolysis se extract karte hain, kyunki inki oxygen ke saath affinity carbon se zyada hoti hai). [Q1086, Q1151]